

1. Given a binary array `nums`, you should delete one element from it. Return the size of the longest non-empty subarray containing only 1's in the resulting array. Return 0 if there is no such subarray.

Example 1:

Input: `nums = [1,1,0,1]`

Output: 3

Explanation: After deleting the number in position 2, `[1,1,1]` contains 3 numbers with value of 1's.

Example 2:

Input: `nums = [0,1,1,1,0,1,1,0,1]`

Output: 5

Explanation: After deleting the number in position 4, `[0,1,1,1,1,1,0,1]` longest subarray with value of 1's is `[1,1,1,1,1]`.

Example 3:

Input: `nums = [1,1,1]`

Output: 2

Explanation: You must delete one element.

2. Write a program to print a Right-Angled Pascal's Triangle pattern for a given number of rows `N`.

Example 1:

Input: `N = 5`

Output:

```
1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
```

3. You are given an array `prices` where `prices[i]` is the price of a given stock on the `i`th day. You want to maximize your profit by choosing a single day to buy one stock and choosing a different day in the future to sell that stock.

Return the maximum profit you can achieve from this transaction. If you cannot achieve any profit, return 0.

Example 1:

Input: `prices = [7,1,5,3,6,4]`

Output: 5

Explanation: Buy on day 2 (price = 1) and sell on day 5 (price = 6), profit = 6-1 = 5.

Note that buying on day 2 and selling on day 1 is not allowed because you must buy before you sell.

Example 2:

Input: `prices = [7,6,4,3,1]`

Output: 0

Explanation: In this case, no transactions are done and the max profit = 0.

4. Given an integer array `nums`, return all the triplets `[nums[i], nums[j], nums[k]]` such that `i != j`, `i != k`, and `j != k`, and `nums[i] + nums[j] + nums[k] == 0`.

Notice that the solution set must not contain duplicate triplets.

Example 1:

Input: `nums = [-1,0,1,2,-1,-4]`

Output: `[[-1,-1,2],[-1,0,1]]`

Example 2:

Input: nums = [0,1,1]

Output: []

Example 3:

Input: nums = [0,0,0]

Output: [[0,0,0]]

5. You're given an even number n. If n=4, you have to print the following pattern:

4444

4334

4334

4444

If n=6, then the pattern should be like this:

666666

655556

654456

654456

655556

666666